

Wanyun Ling

📍 Hanbrucher Str. 26, 52064 Aachen, Germany ✉ wanyun.ling@tum.de ☎ +49 159 01346100

Research Interests

Multimodal Spatio-Temporal Data Mining, Time Series Analysis, GeoAI, Interpretable Machine Learning

Education

Technical University of Munich

Apr. 2026 - Present

Doctorate (*Dr. rer. pol.*)

- **Field:** Spatio-Temporal Data Mining, Time Series Analysis, Urban AI
- **Supervisor:** Prof. Dr. Ziyue Li **Mentor:** Dr. Chenxi Liu
- Full-time — 1st year

RWTH Aachen University

Oct. 2023 - Feb. 2026

Master of Science in Transforming City Regions

- GPA: 1.7 (German grading system, *Gut/Good*; Top 5%)
- **Core Courses:** Urban Transformation, Territorial Analysis and Digital Tools (GIS), Research Methods, Urban Policies
- **Achievements:** Achieved top marks in several core modules; selected coursework was featured in the RWTH Städtebau Workflow university publication
- **Awards:** STIBET I Graduation Scholarship, RWTH Aachen University (Feb.–Apr. 2026)

Anhui Jianzhu University

Sept. 2018 - Jun. 2023

Bachelor of Architecture

- GPA: 3.48/5.0 (rank: top 10%)
- **Core Courses:** Urban Development and Planning Control, Traffic Planning and Management, Urban Sociology, Behavior and Environment, C Programming, Parametric Design
- **Awards:** Outstanding Graduate Scholarship, Anhui Jianzhu University
- **Thesis:** Research on the Path Design of Parks in the Old City of Hefei Based on Slime Mold Behavior Simulation and Optimization (Supervisor: A.P. Jieqing Xu, Jan. 2023 - Jun. 2023)

Papers & Conferences

Atmosphere-Aware Place Discovery: Semantic Urban Mapping with Multimodal Vibe Enrichment [🔗](#)

Under Review

Mikhail Aizinov, **Wanyun Ling**, Evgenii Kniazev, Chenxi Liu, **Ziyue Li**

ACM SIGSPATIAL '26, ACM, New York, NY, USA (Under Review)

Urban Climate Justice through Interpretable Ensemble Learning: Heat Exposure Inequities and Planning Strategies in Metro Vancouver [🔗](#)

Dec 2025

Wanyun Ling

61st ISOCARP World Planning Congress 2025, Riyadh, Saudi Arabia, December 1–4, 2025.

Adaptive Urban Heat Mitigation Through Ensemble Learning: Socio-Spatial Modeling and Intervention Analysis [🔗](#)

Oct 2025

Wanyun Ling^{1*}, **Liyang Chu**²

Buildings, 15(21), 3820. <https://doi.org/10.3390/buildings15213820> [🔗](#) (IF 3.2; JCR Q2)

Data-Driven Circularity: Visualizing Urban Heat Inequity for Just Green Transitions – A Case Study in Vancouver [🔗](#)

Sep 2025

Wanyun Ling

Urban Futures Symposium 2025, Dübendorf, Switzerland, September 1–2, 2025

Ensemble Learning Approaches for Adaptive Urban Heat Mitigation: A Socio-Spatial Analysis [🔗](#)

Aug 2025

Wanyun Ling

Global Smart Cities Summit & The 4th International Conference on Urban Informatics (GSCS & ICUI 2025), Hong Kong, August 5–8, 2025

Flut der Zeit: Aufbau einer lebendigen Nachbarschaft für verschiedene Altersgruppen [🔗](#)

2024

Wanyun Ling, Yunyi Huang

Städtebau Workflow: Kupferstadt Stolberg, RWTH Aachen University, pp. 36–39

Technical Skills

Languages: English (proficient), German (proficient), Mandarin (native)

Programming & Data Analysis: Python (proficient), R, SQL, Machine Learning (scikit-learn), Deep Learning (PyTorch), Statistical Analysis, Time Series Analysis

Geospatial & Urban Analysis: QGIS, ArcGIS, Remote Sensing, Satellite Image Analysis, Street View Data Processing, Spatial Statistics

Honors & Awards

STIBET I Graduation Scholarship, RWTH Aachen University — Feb.–Apr. 2026

Outstanding Graduate Scholarship, Anhui Jianzhu University — 2023

Third-Class Scholarship, Anhui Jianzhu University — Awarded consecutively from 2018 to 2023

Research Projects

Urban Heat Inequality and Adaptive Mitigation: Interpretable Machine Learning for Socio-Spatial Analysis in European Cities [🔗](#)

Sept. 2025 - Feb. 2026

Master Thesis, RWTH Aachen University, Faculty of Architecture

- **Supervisors:** Univ.-Prof. Dr. Jakob Beetz; Prof. Verónica García Donoso
- **Objectives:** Examined socio-spatial inequalities in urban heat exposure across Zuid-Holland and evaluated how alternative UHI mitigation strategies affect cooling performance, social equity, and gentrification risk.
- **Methodologies:** Developed a 500-m grid-based spatial dataset combining satellite-derived UHI, NDVI, urban morphology, blue-green infrastructure, and socio-demographic indicators; applied ensemble AutoML, feature-importance analysis, scenario simulation, vulnerability grouping, property-value modeling, and Bayesian multi-objective optimization.
- **Outcomes:** Identified key drivers of daytime and nighttime UHI exposure; demonstrated that vegetation and water-related interventions provide strong cooling but may generate housing-market pressures; proposed optimized urban-form interventions that reduce heat while limiting gentrification risk.

Pedal to a Greener Future: Pioneering Zuid-Limburg's Bike-Friendly Infrastructure [🔗](#)

Apr. 2024 - Jul. 2024

RWTH Aachen University

- **Supervisor:** Univ.-Prof. Dipl.-Ing. Christa Reicher
- **Objectives:** Develop phased strategies for sustainable bicycle infrastructure in Zuid-Limburg, enhancing rural connectivity and regional socio-economic resilience
- **Methodologies:** Utilized transition management and adaptive governance frameworks; conducted stakeholder workshops, demographic analysis, and scenario modeling; Applied GIS-based spatial analysis for route optimization
- **Outcomes:** Proposed phased implementation scenarios and formulated a strategic model integrating cultural-historical elements; Identified optimal routes considering environmental factors and user preferences

Waterwise Urbanism: Integrating Solutions from Regions to Neighbor-

Apr. 2024 - Jul. 2024

- **Supervisor:** Prof. Verônica Garcia Donoso
- **Objectives:** Developed multi-scalar strategies to address flood and erosion risks in the Maas Basin, enhancing climate resilience while balancing ecological sustainability and community well-being.
- **Methodologies:** Conducted climate, hydrological, and vegetation analyses; applied nature-based solutions (e.g., retention basins, bioswales, rain gardens, permeable pavement) across regional, city, and neighbourhood scales.
- **Outcomes:** Proposed an integrated framework for "waterwise" urbanism that improved flood resilience and biodiversity; selected work was featured in the RWTH Städtebau Workflow publication.

Internship Experience

Assistant of Community Planning

Aug. 2022 - Dec. 2022

Hefei Natural Resources and Planning Information Center

- **Project:** Longqiao Town, Lujiang County Community Planning
- Integrated GIS analysis, stakeholder interviews, and literature review to assess land use and community needs
- Completed the basic design of a historical village museum and project drawings
- Advanced community understanding of land use dynamics and delivered a museum design respecting cultural legacy

Assistant of Residential Plan

Jun. 2022 - Aug. 2022

Hefei Planning and Design Institute

- **Project:** Luyang District Residential Plan
- Conducted site study and local culture and history research
- Utilized spatial-environmental analysis and passive design strategies for ventilation and solar access evaluation
- Enhanced residential plan with data-driven insights and produced detailed project drawings and renderings